# 8-5-1: Examining IPv6 Addresses

At the end of this episode, I will be able to:

1. Identify and explain the characteristic of IPv6, given a scenario.

Learner Objective: *Identify and explain the characteristic of IPv6, given a scenario*

Description: In this episode, the learner will examine IPv6 addresses, inlcuding the format and structure. We will explore the benefits and characteristics of IPv6.

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- What is an IPv6 address
    - Utilizes 128-bit addresses, enabling an extensive number of unique IP addresses.
    - Portions of the address are reserved for the network prefix (like an IPv4 network ID)
    - The remaining portion is reserved for the interface identifier (like an IPv4 host ID)
- What are some of benefits of IPv6
    - Vast Address Pool
        - The large address space removes the need for NAT, simplifying network design.
    - Simplified Routing and Header Format
        - Hierarchical structuring for efficient routing; simplified packet header.
    - Elimination of Broadcast Communications
        - Replaces broadcasting with multicast and anycast.
    - Stateless and Stateful Configuration
        - Supports both automatic (SLAAC) and dynamic (DHCPv6) IP addressing.
    - Inherent Security Features
        - Designed with IPsec for enhanced security.
    - Enhanced Support for Mobile Nodes
        - Mobile IPv6 reduces latency for mobile connectivity.
- What are the different types of IPv6 address?
    - Different IPv6 Address Types
        - Global Unicast Addresses (Prefix- 2000::/3): Unique and routable globally.
        - Unique Local Addresses (Prefix- FC00::/7): For local communications, not globally routable.
        - Link-Local Addresses (Prefix- FE80::/10): For communications on a single network link.
        - Multicast Addresses (Prefix- FF00::/8): For one-to-many communication.
        - Anycast Addresses: Assigned to multiple interfaces, with the nearest one responding.
        - Special Addresses- Teredo (Prefix- 2001::/32), 6to4 (Prefix- 2002::/16), ISATAP.
- What is EUI-64?
    - Converts a 48-bit MAC address to a 64-bit interface identifier for IPv6.
    - EUI-64 Format
        - Extends a 48-bit MAC address to 64 bits for IPv6 interface identification.
        - Involves inserting 'FFFE' in the middle of the MAC address and inverting the 7th bit (Universal/Local bit).
- What is SLAAC?
    - StateLess Address AutoConfiguration, is a method used in IPv6 networking to enable devices on a network to automatically configure their own IPv6 addresses without the need for a server-based addressing mechanism like DHCP.

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- Additional Resources:

- If applicable